

LAB REPORT
FOR
LAB-6: OFDM MODULATION & FREQUENCY DOMAIN
EQUALIZATION

EEC 556: SOFTWARE DEFINED RADIO

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INTRODUCTION

In this lab we will implement the key features of the orthogonal frequency division multiplexing (OFDM) multicarrier modulation technique. OFDM is a transmission technique with a special structure that permits low complexity linear equalization at the receiver. Several commercial wireless systems have adopted OFDM modulation including wireless LAN standards like IEEE 802.11g, IEEE 802.11a, and IEEE 802.11n; broadband wireless access including IEEE 802.16 (Wi Fi); mobile broadband wireless known as IEEE 802.20; digital video broadcasting DVB (used in Europe); as well as several releases of the 3GPP cellular standards.

Although the design of an OFDM system has several important differences from the single carrier system considered in previous labs, it must still perform many of the same functions (e.g., channel estimation, equalization, and frequency offset estimation). In this lab we will still perform all functions related to synchronization and channel estimation in the time domain using the same training sequence as discussed in Lab 5.

BACKGROUND

OFDM is a type of digital modulation where information is modulated into discrete-time sinusoids. With OFDM, the symbols after the constellation mapping are considered to start in the frequency do-main. OFDM operates on groups of symbols, the resulting group of symbols being called an OFDM symbol.

In this lab we will implement two lab virtual instruments:

1. **OFDM MODULATOR.VI**
2. **OFDM DEMODULATOR.VI**

We will be using a set of helper VIs which have been provided in OFDM_comm1.0.1lb, a new digital communications library for processing related to OFDM.

1. OFDM MODULATOR.VI:

We have designed the ofdm_modulator.vi using lab manual and lab notes as shown in figure 1.

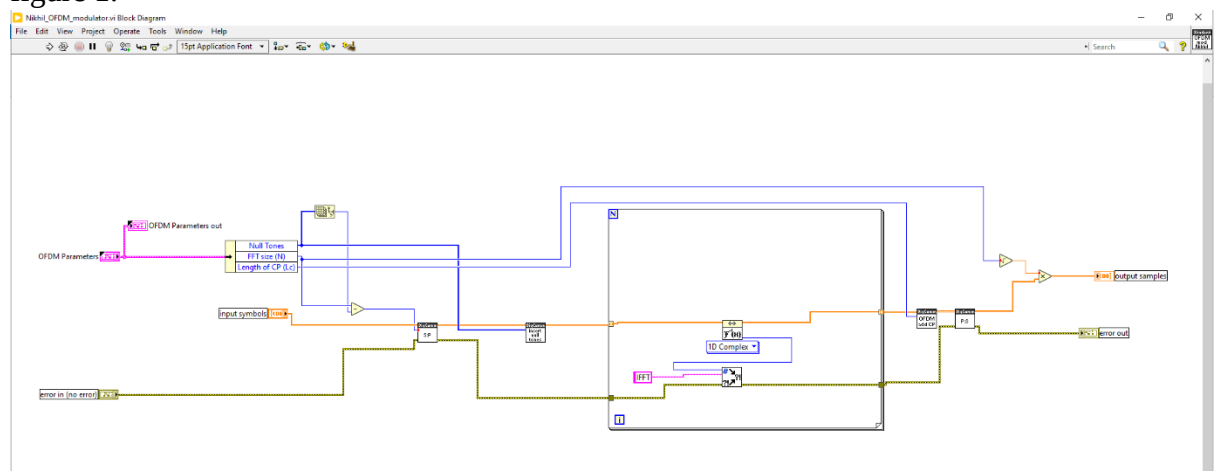


Figure 1: Nikhil_OFDM_modulator.vi

2. OFDM DEMODULATOR.VI:

We have designed the ofdm_demodulator.vi using lab manual and lab notes as shown in figure 2.

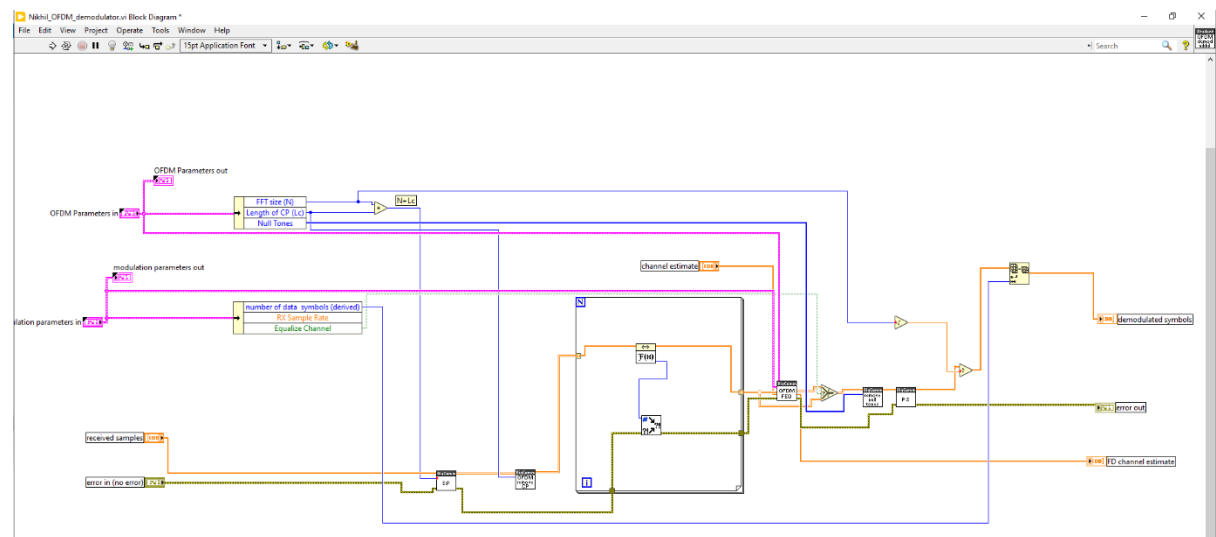


Figure 2: Nikhil_OFDM_demodulator.vi

SIMULATION:

We have replaced both the vi's ofdm_modulator.vi & ofdm_demodulator.vi in transmitter.vi & receiver.vi as shown in figure 3&4 respectively.

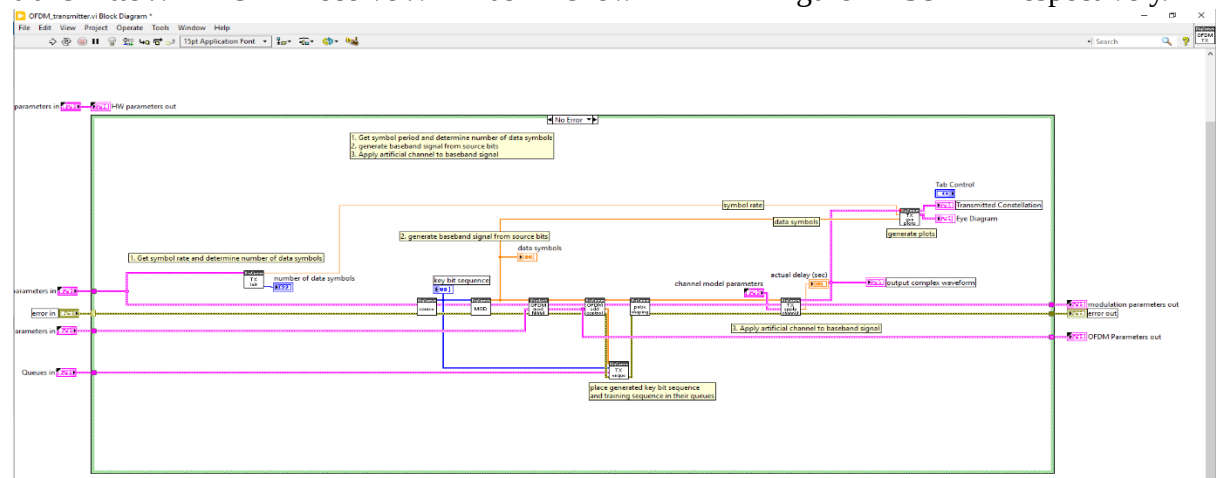


Figure 3: OFDM_transmitter.vi (replaced with modulator)

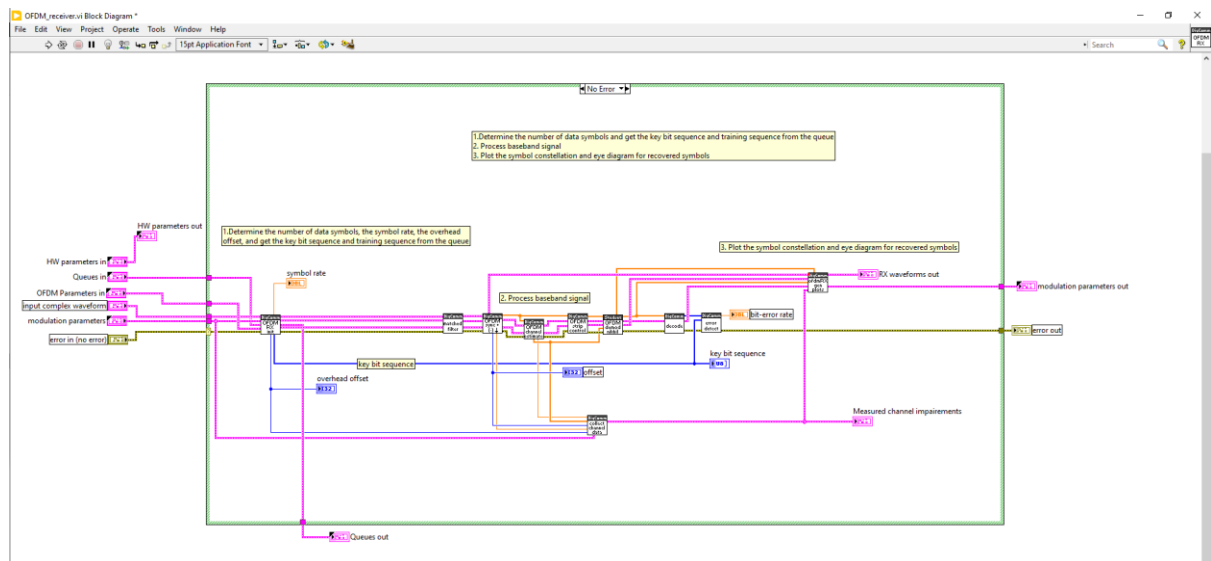


Figure 4: OFDM_receiver.vi(replaced with demodulator)

We observed simulation results on simulator.vi as shown in figures 5,6,7.

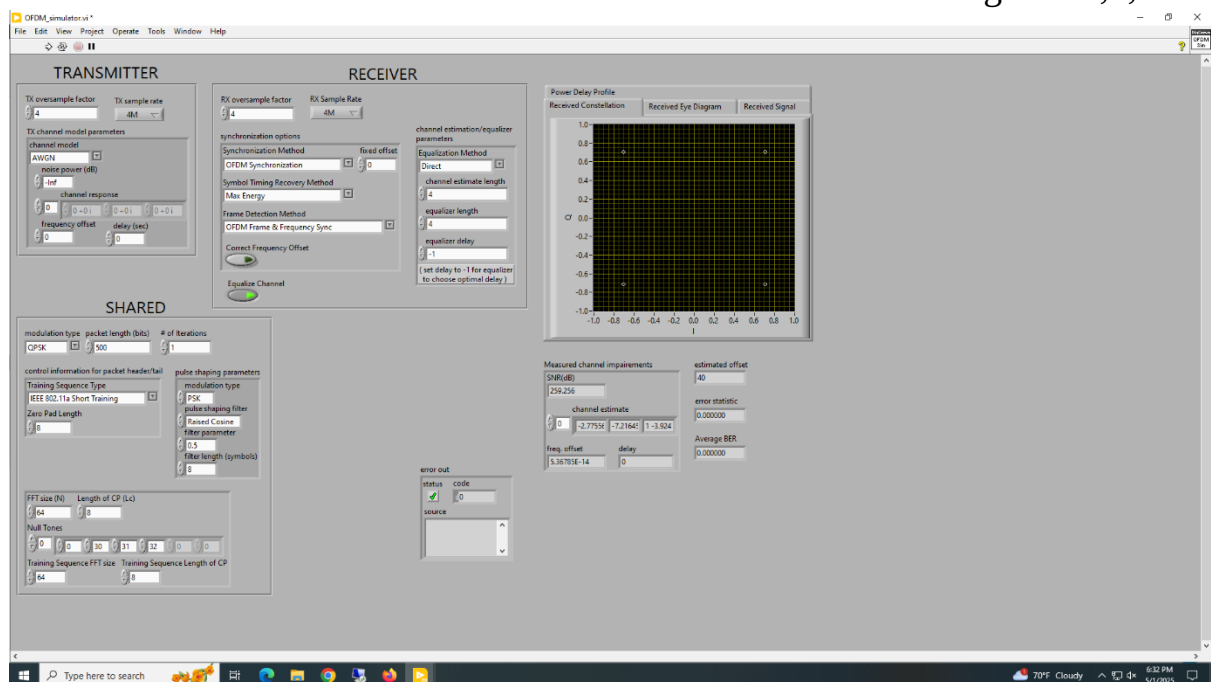


Figure 5: OFDM_simulator.vi(Received constellation)

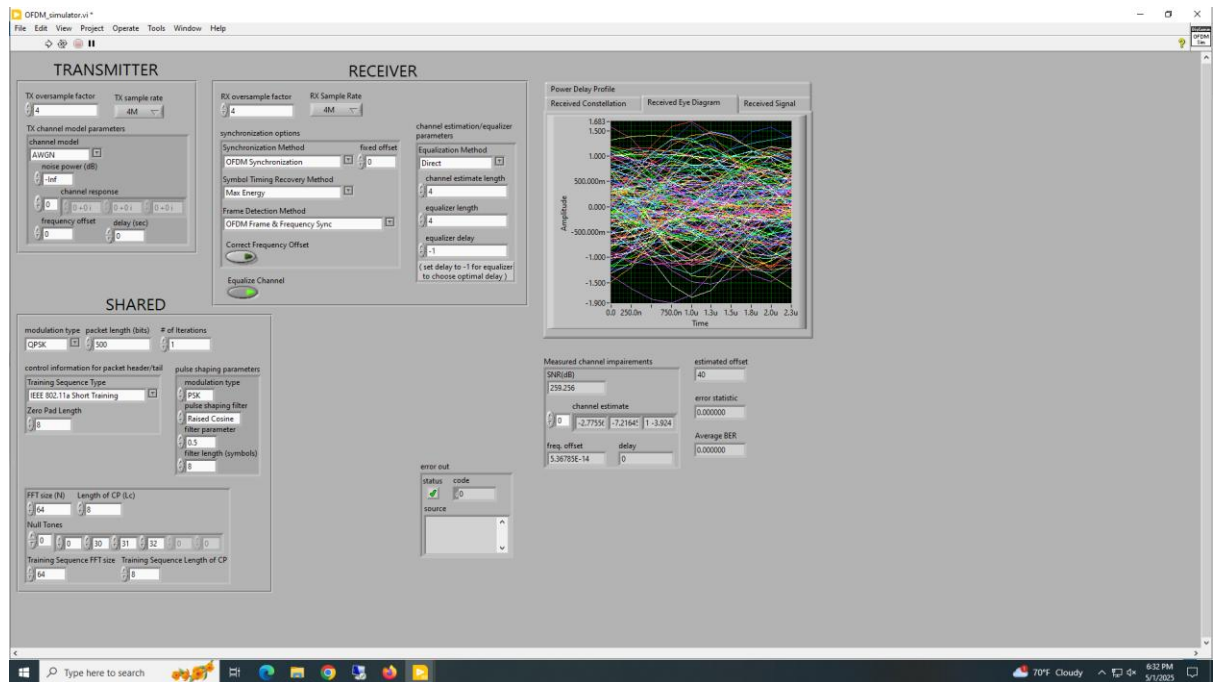


Figure 6: OFDM_simulator.vi(Received eye diagram)

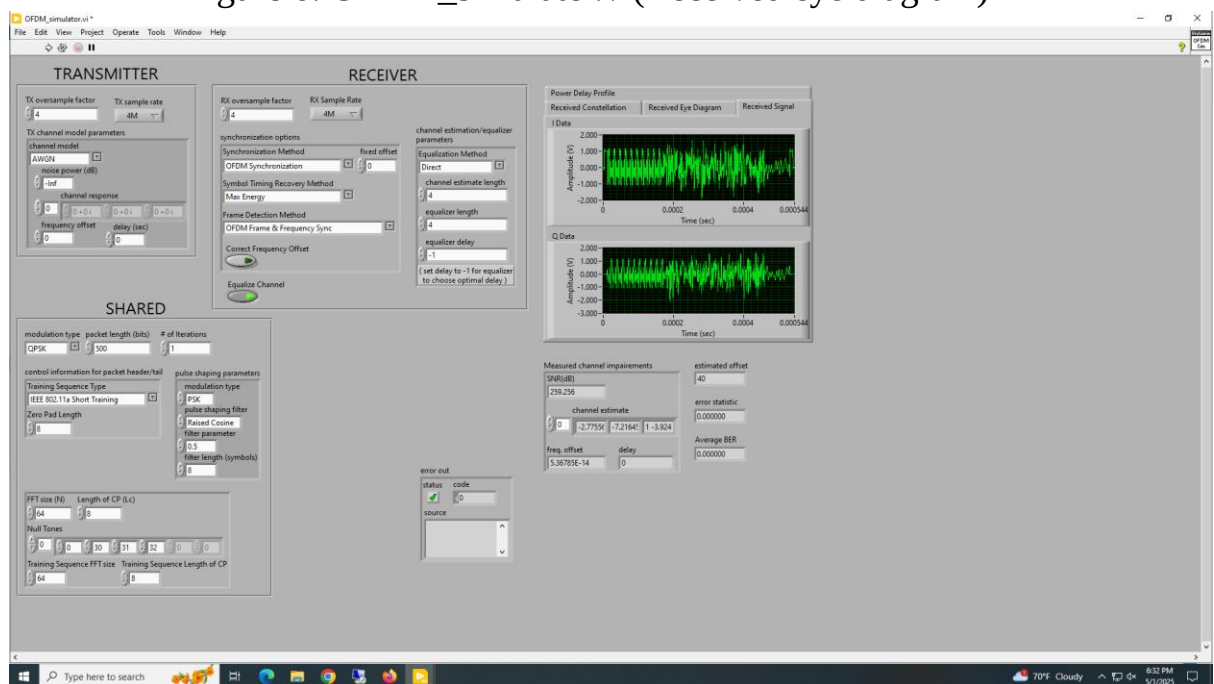


Figure 7: OFDM_simulator.vi(Received signal)

LAB EXPERIMENT

We Set up the parameters listed below in top_tx.vi and top_rx.vi. we observed results as shown in figures 8,9,10.

- Packet length = 500 bits
- Modulation type = QPSK
- Channel estimate length = 4
- FFT size (N) = 64
- Length of CP (L_c)=8
- Null tones = {0, 31, 32, 33}

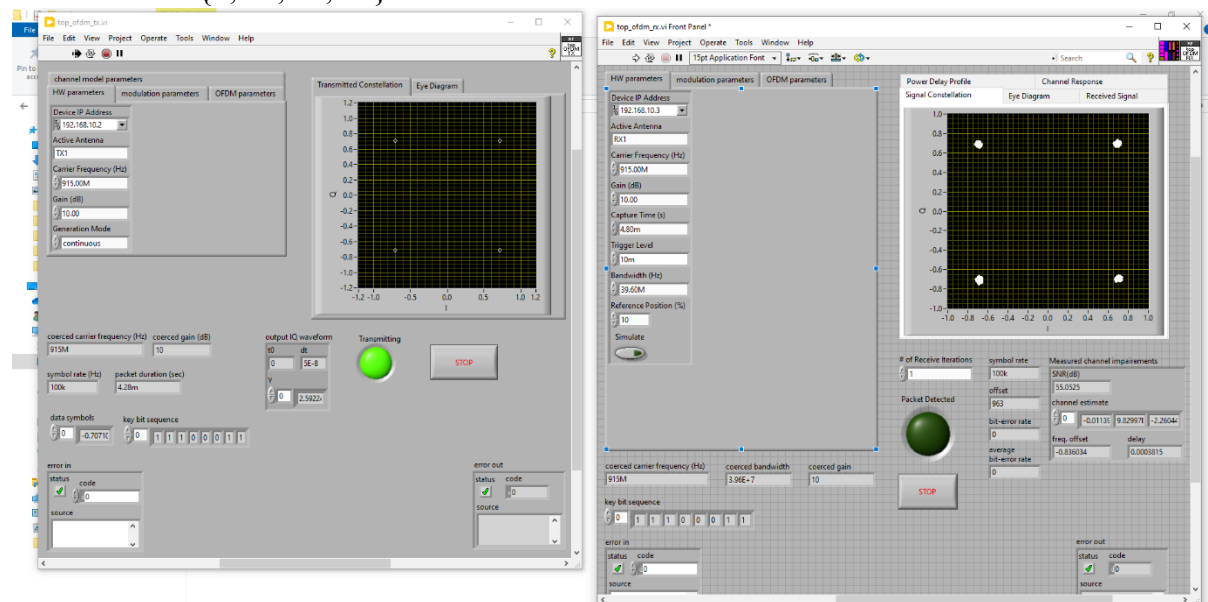


Figure 8: top_tx.vi & top_rx.vi(signal constellation)

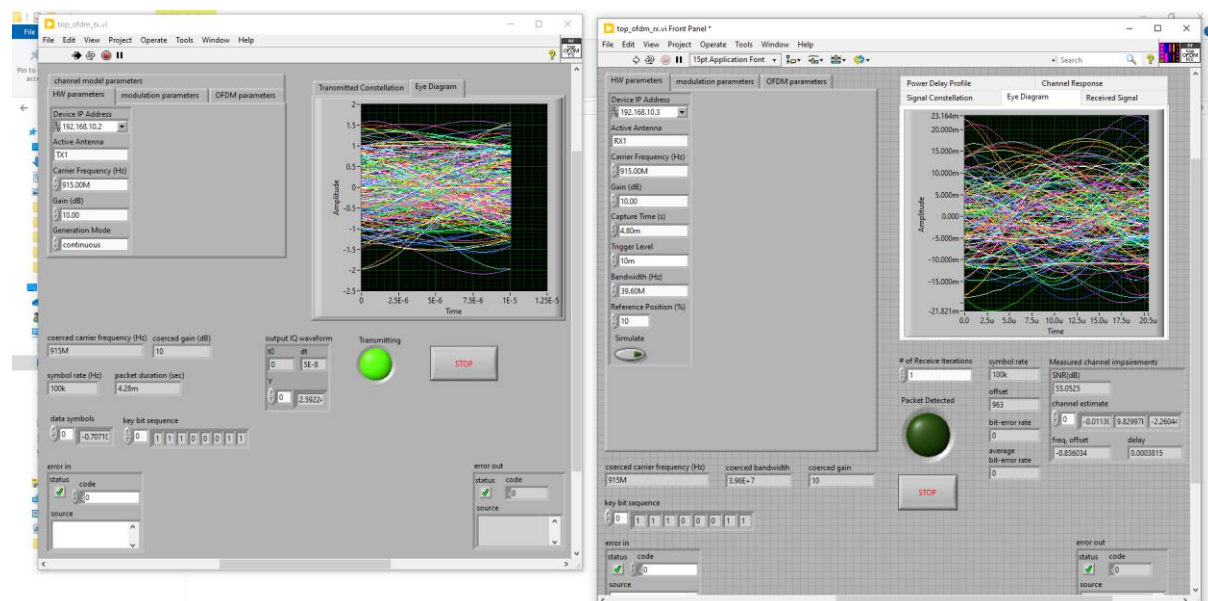


Figure 9: top_tx.vi & top_rx.vi(eye diagram)

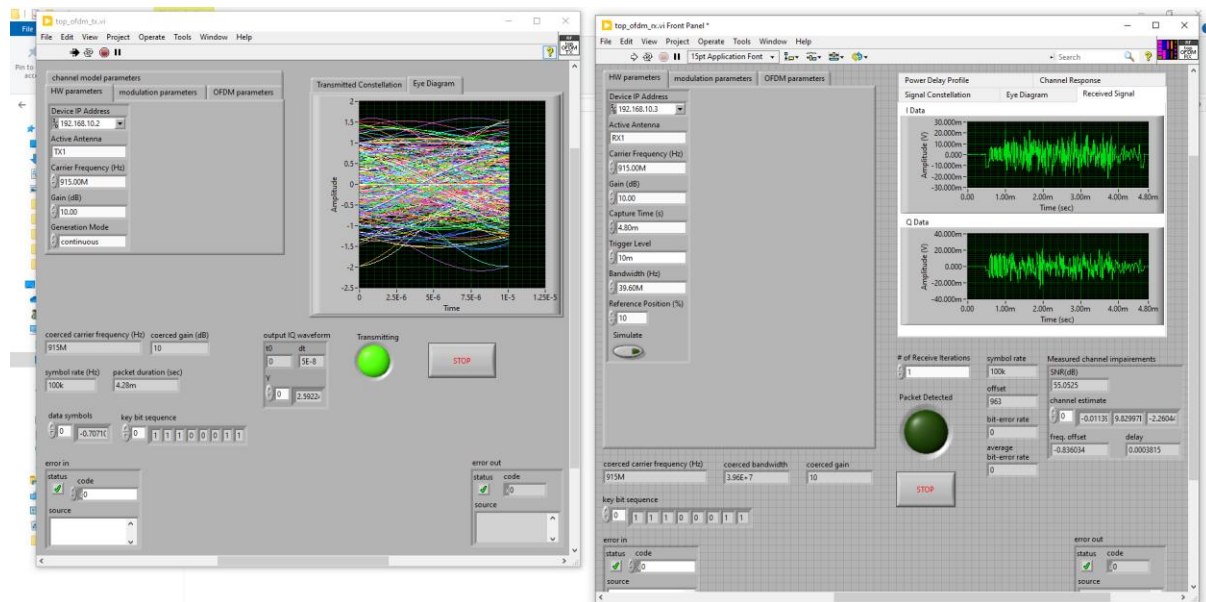


Figure 10: top_tx.vi & top_rx.vi(Received signal)

3.2 Frequency Selectivity of wireless channels

Narrowband: we have set the following parameters on top_tx.vi & top_rx.vi as shown in figures 11,12,13.

- TX sample rate = 4 MSamples/sec
- TX oversample factor = 20
- RX sample rate = 4 MSamples/sec
- RX oversample factor = 20
- Capture time = 2.4 msec

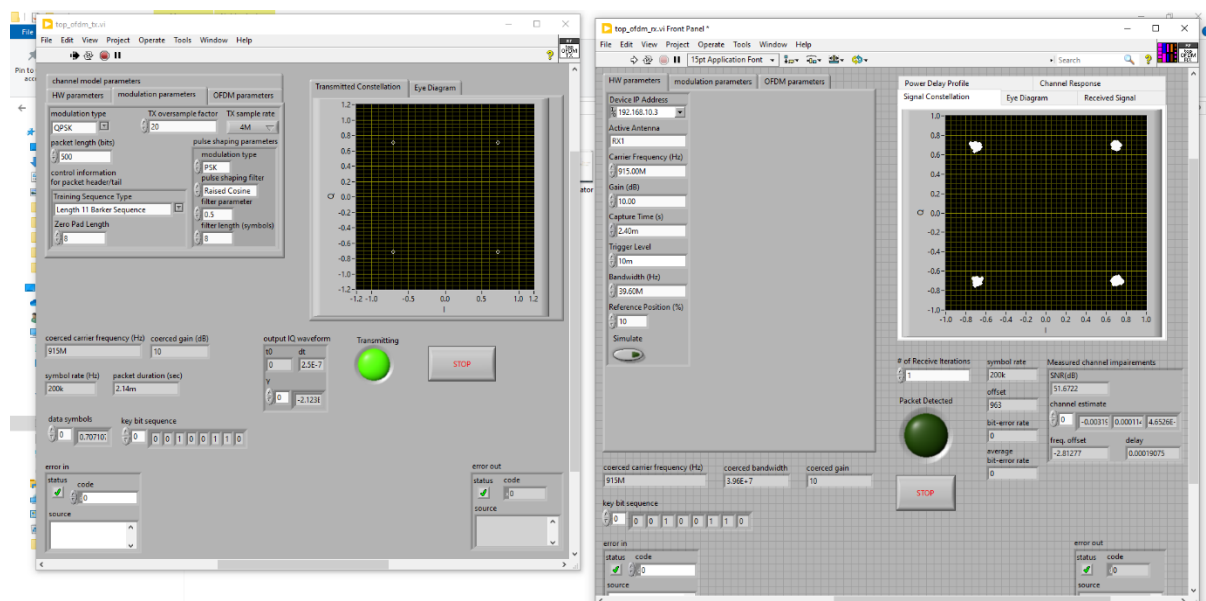


Figure 11: top_tx.vi & top_rx.vi(signal constellation)

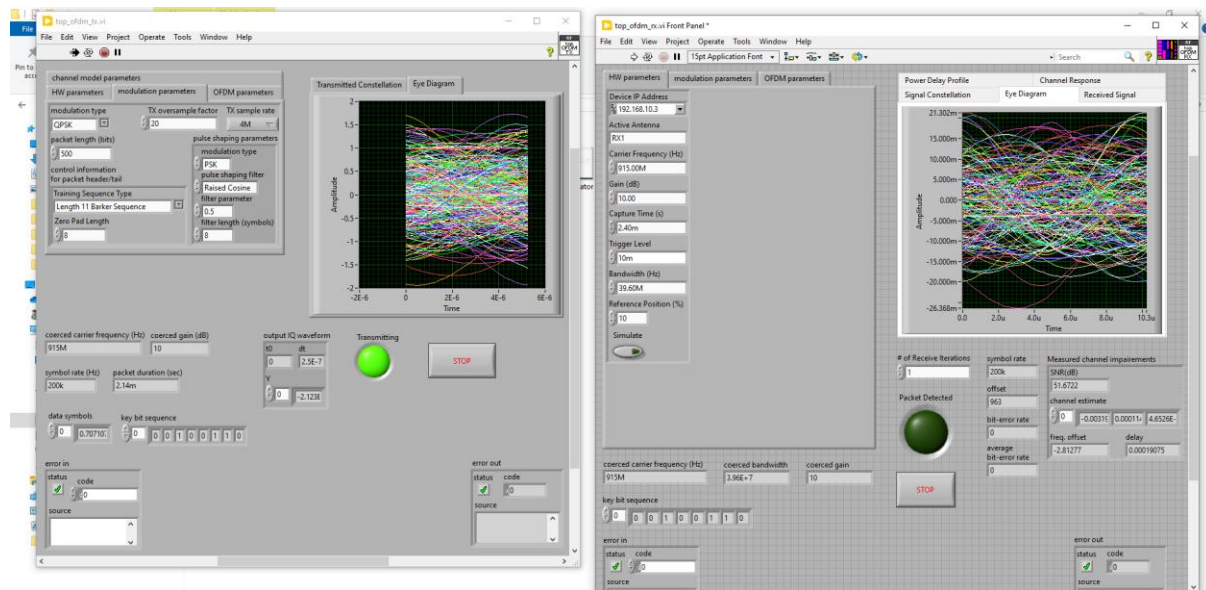


Figure 12: top_tx.vi & top_rx.vi(Eye diagram)

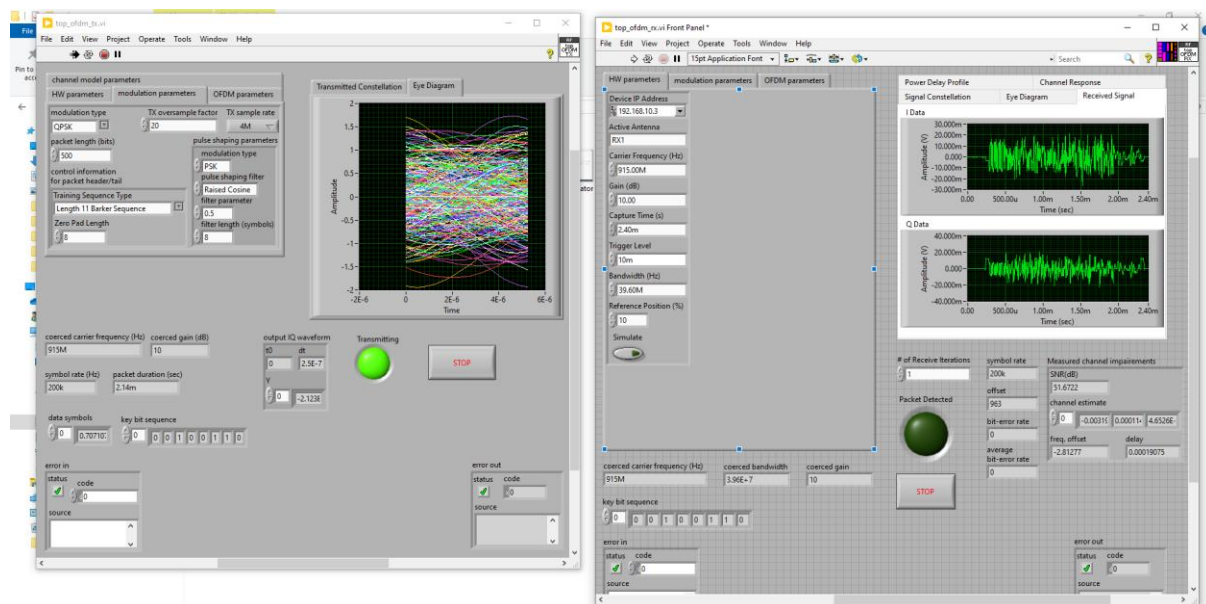


Figure 13: top_tx.vi & top_rx.vi(Received signal)

Wideband: Set the following parameters in top_tx.vi & top_rx.vi as shown in figures 14,15,16.

- TX sample rate = 20 MSamples/sec
- TX oversample factor = 4
- RX sample rate = 10 MSamples/sec
- RX oversample factor = 2
- Capture time = 100 μ sec

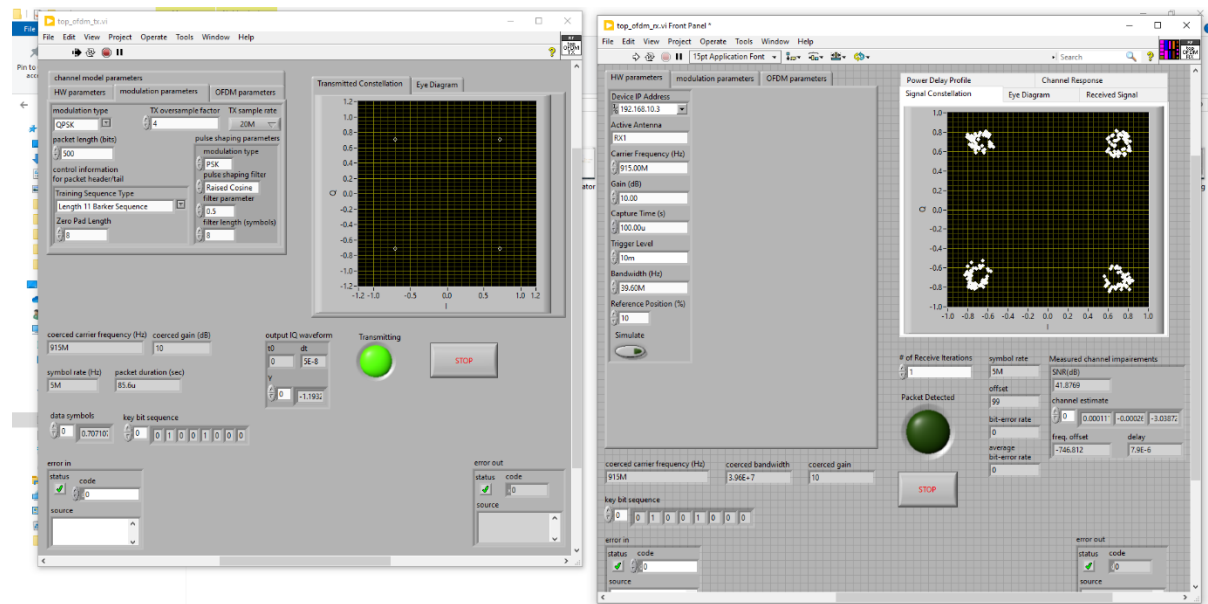


Figure 14: top_tx.vi & top_rx.vi(signal constellation)

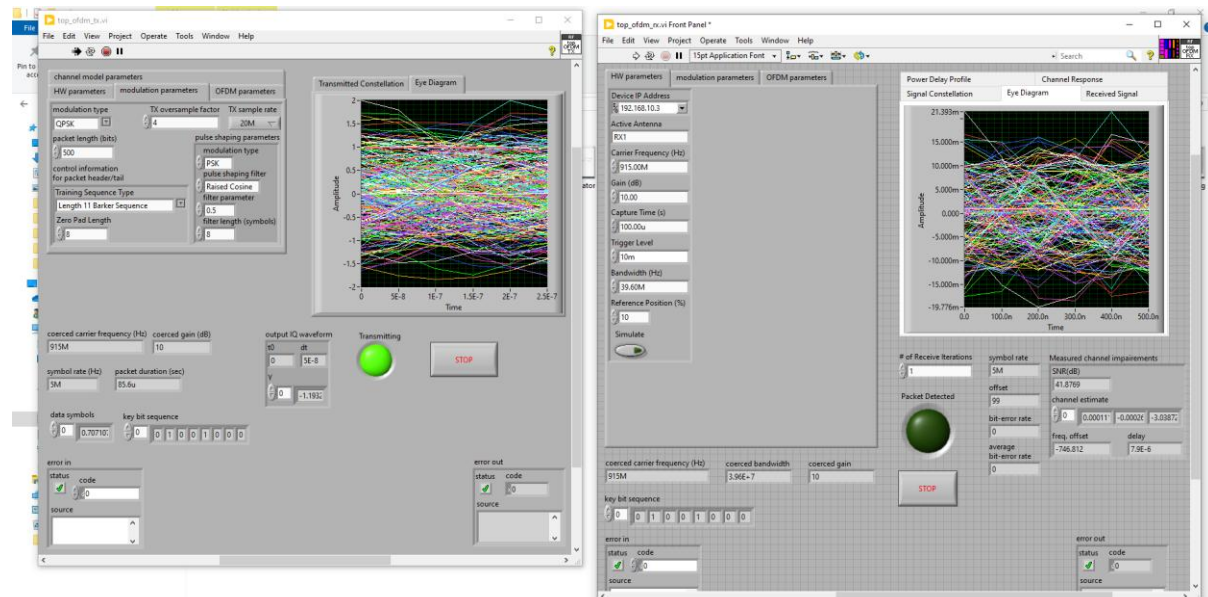


Figure 15: top_tx.vi & top_rx.vi(Eye diagram)

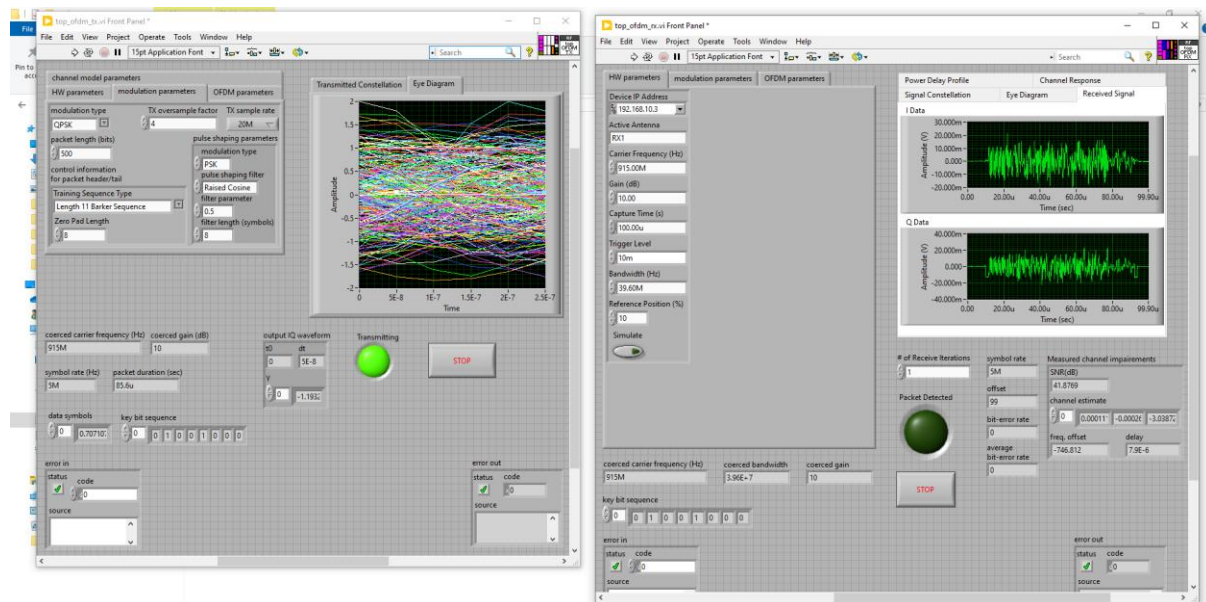


Figure 16: top_tx.vi & top_rx.vi(Received signal)

Questions and answers

1. According to the parameters above, what is the respective OFDM symbol rate in the narrowband and wideband systems you have set up (i.e.. the reciprocal of the OFDM symbol period as previously defined)?

Ans: Symbol rate for OFDM system given by $\frac{1}{T(N+Lc)} = \frac{\text{sample rate}}{(N+Lc)}$

For narrowband: $\frac{4}{(64+8)} = 55K \text{ samples/sec}$

For wideband: $\frac{20}{(64+8)} = 277K \text{ samples/sec}$

2. What are the effective lengths of the narrowband and wideband channels respectively?

Ans: Effective lengths for narrowband channel is 1 and wideband channel is 2.

3. Describe the frequency responses of each channel. In particular, are the frequency responses of these channels frequency selective or flat?

Ans: The frequency response of both channels exhibits frequency selectivity due to null tones. Multicarrier systems utilize a cyclic prefix to facilitate digital frequency domain equalization. Orthogonal Frequency Division Multiplexing (OFDM) segments a frequency selective channel into N flat fading subchannels, allowing OFDM to be viewed as N parallel flat fading channels, (each with a bandwidth of $1/NT$), multiplexed in the frequency domain. Consequently, a straightforward zero-forcing equalizer can be applied to each subchannel to equalize channel impairments.

2.3 sensitivity to frequency offsets

In this experiment we will observe how the performance of an OFDM system degrades in the presence of a frequency offset. Setting up the following parameters in our system.

- TX sample rate = 20 MSamples/sec
- TX oversample factor = 20
- RX sample rate = 4 MSamples/sec
- RX oversample factor = 4
- Capture time = 500 μ sec
- Frequency offset (Hz) = 200 Hz

To observe how a frequency offset impacts our system, we will need to disable the frequency offset correction algorithm implemented in the synchronization block. To do this, simply set Correct Frequency Offset on the front panel of top OFDM_rx.vi to FALSE. Now you will consider two OFDM systems which use different values for N. First, consider a system which uses $N = 64$ subcarriers.

Set up the following OFDM parameters.

- FFT size (N) = 64
- Length of CP (L_c) = 8
- Null tones = {0, 31, 32, 33}

And observed results as shown in figures 17,18.

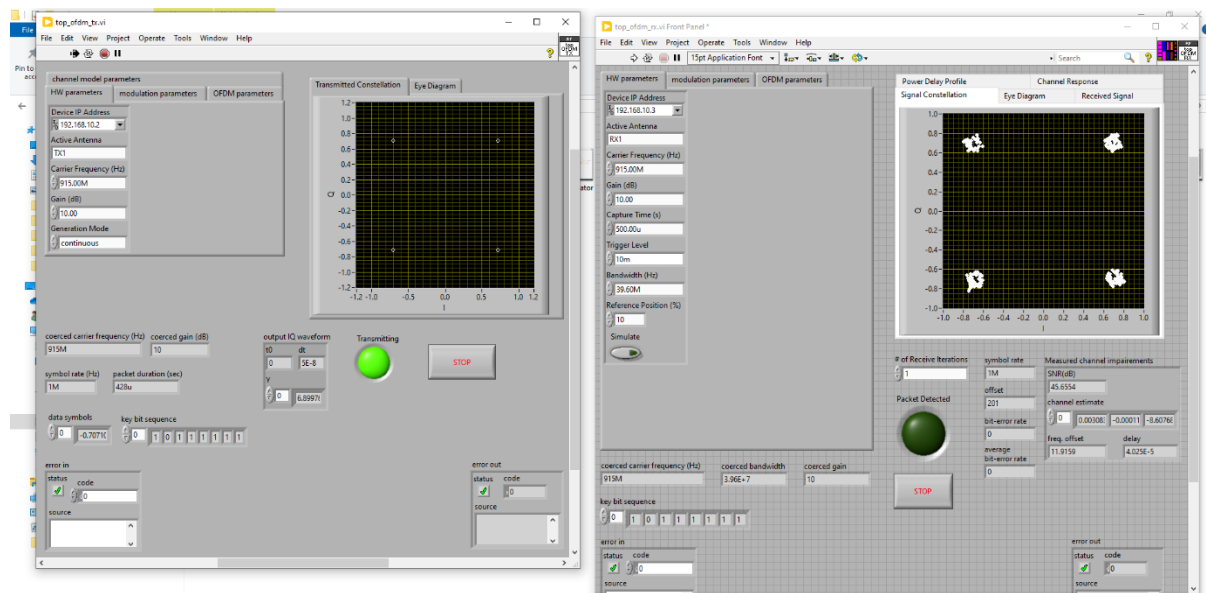


Figure 17: top_tx.vi & top_rx.vi(signal constellation)

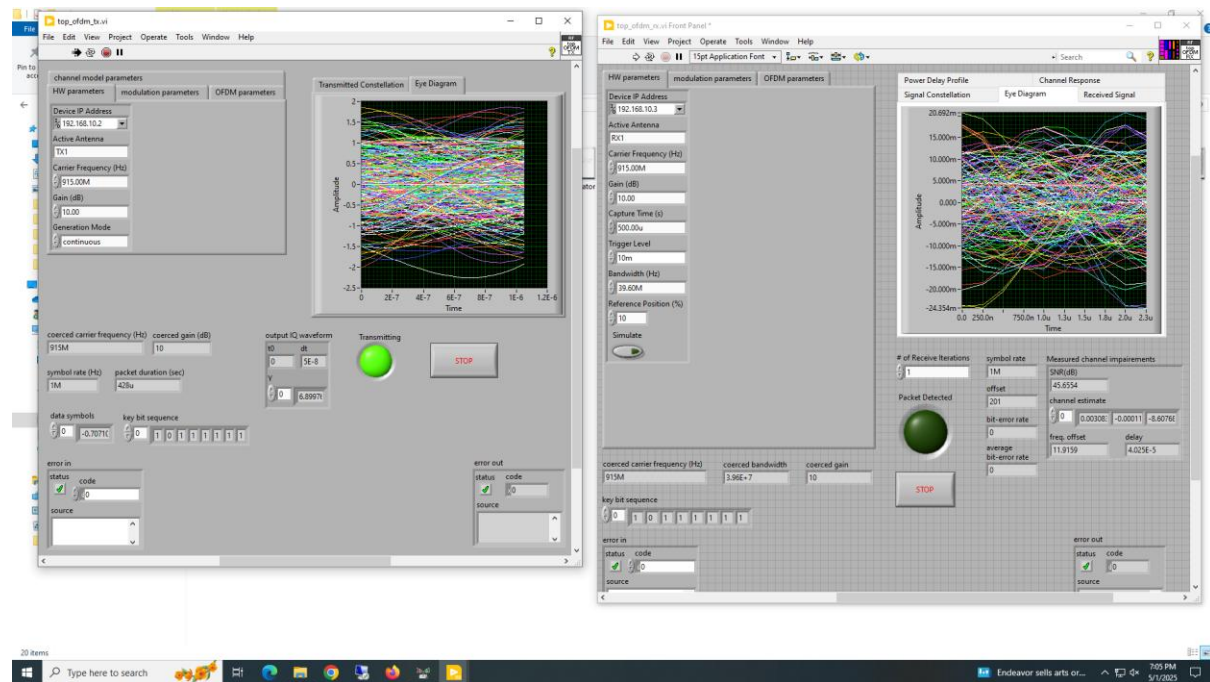


Figure 18: top_tx.vi & top_rx.vi(eye diagram)

Now consider a system which uses $N = 1024$ subcarriers.

- FFT size (N) = 1024
- Length of CP (L_c) = 32
- Null tones = $\{0, 511, 512, 513\}$

And observed results in figures 19,20,21.

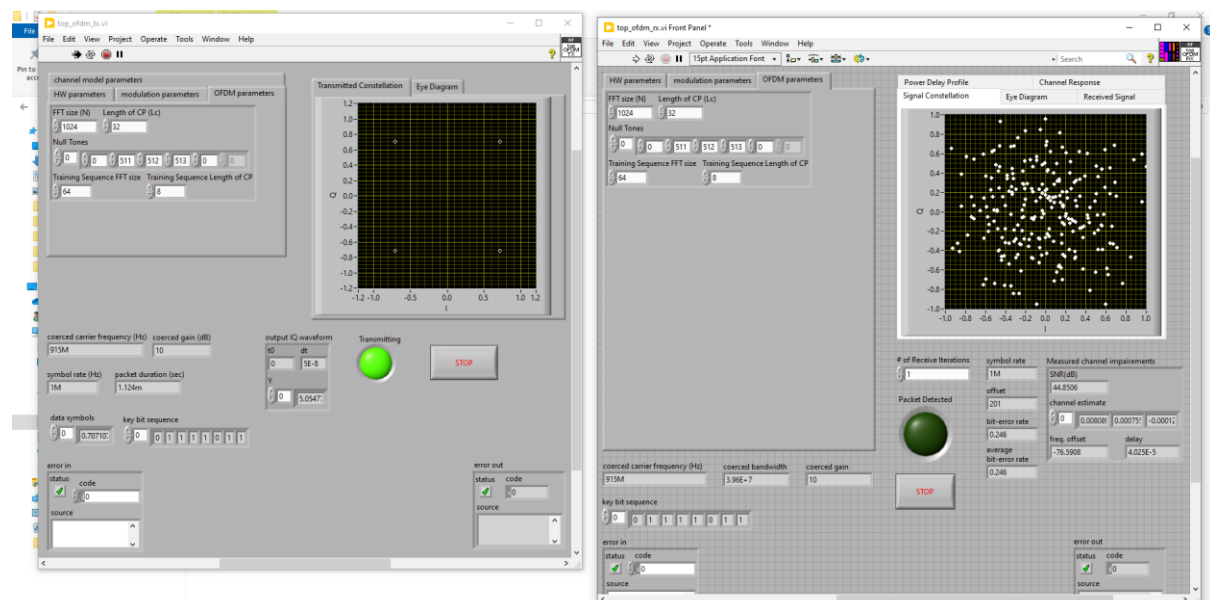


Figure 19: top_tx.vi & top_rx.vi(signal constellation)

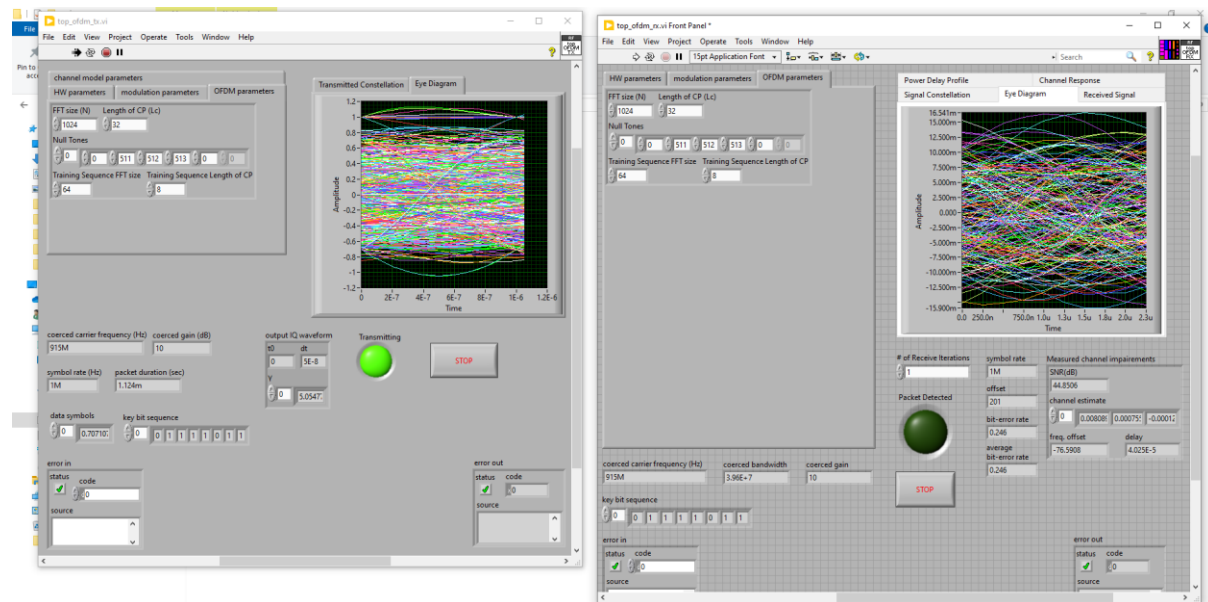


Figure 20: top_tx.vi & top_rx.vi(eye diagram)

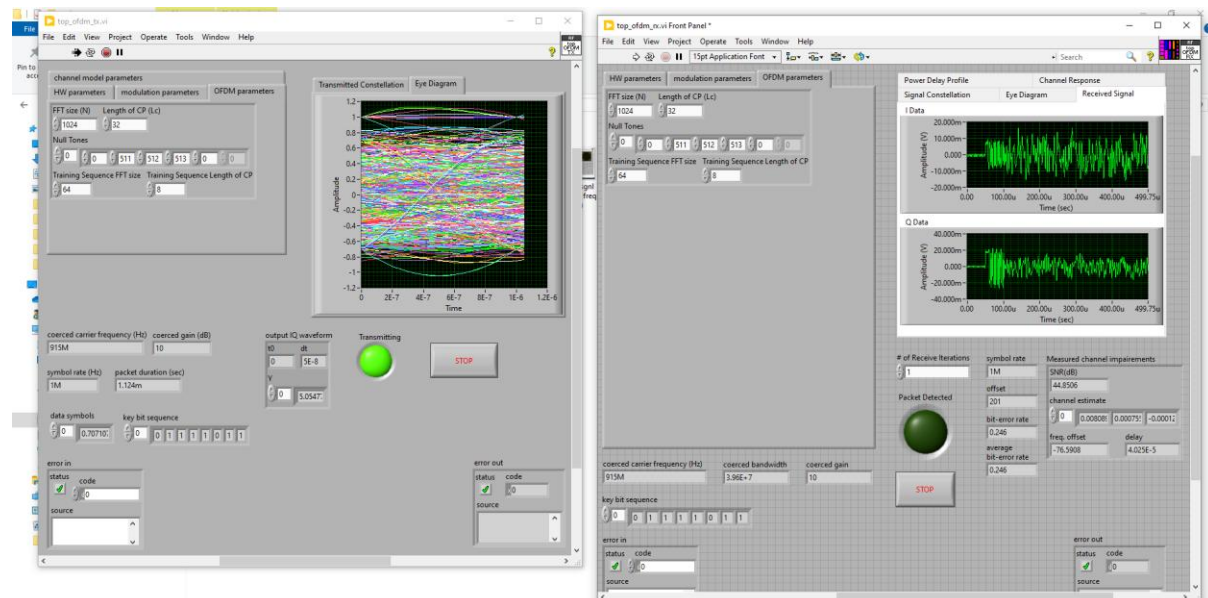


Figure 21: top_tx.vi & top_rx.vi(Received signal)

Question and answers:

1. Recall from the Lab 5 that in a single carrier system, a frequency offset causes a time varying phase offset which effectively "smears" the received constellation as it rotates it. How is the impact of a frequency offset in OFDM systems different from that of single carrier systems? In particular, how is the impact on the signal constellation different?

Ans: In a signal carrier system, the entire signal becomes different as it rotates due to frequency offset. In contrast, with OFDM, the individual effects can be observed in the constellation of each signal element.

2. What is the subcarrier spacing Δ_c of your system when $N = 1024$ and 64 respectively?

Ans: Subcarrier spacing is given by $\Delta_c = \frac{BW}{N} = \frac{1}{NT}$; where $\frac{1}{T} = \text{sample rate} = 20M \text{ samples/sec}$

For $N = 1024$, $\Delta_c = \frac{20M}{1024} = 19.531K \text{ Hz}$

For $N = 64$, $\Delta_c = \frac{20M}{64} = 312.5k \text{ Hz}$

3. Which of the systems (i.e., $N = 1024$ or 64) is more severely impacted by a 200 Hz frequency offset? Why?

Ans: System with $N = 1024$ is more severely impacted by a 200 Hz frequency. From the above calculation of subcarrier spacing, higher the subcarrier spacing higher the Bandwidth and Sample rate, hence the resolution of the system. As so higher the value of N lowers the subcarrier spacing.

4. As mentioned, frequency offsets can be interpreted as shifts in the frequency domain. There is a duality between the effect of this shifting in a multicarrier system and symbol timing error in a single carrier system (discussed in Part 2 of Lab 2).

Discuss this relationship between the effect of symbol timing error in single carrier systems and frequency offset in OFDM systems. Be sure to include in your discussion the effect of each impairment on the received signal constellation. Hint: Think about the impact of each type of error on the orthogonality (or separation) between symbols.

Ans: When it comes to symbol timing errors, both signal carrier systems and OFDM face challenges that can lead to poor performance. In a signal carrier system, a timing error causes the received signal to shift and become distorted, resulting in inter-symbol interference (ISI). Similarly, in an OFDM system, a timing error leads to inter-carrier interference (ICI), which distorts the subcarriers and introduces noise. This ultimately affects the quality of the received signal, resulting in reduced overall performance for both systems.

Question and answers:**2. Discuss any problems you might have encountered and how you over-came these obstacles.**

Ans: we have encountered the problem with OFDM_demodulator.vi and overcome this problem with lab notes and professor inputs.

3. In this lab you observed how an OFDM system operates over a real wireless link. Answer the following questions about your experience in lab.

(a) OFDM systems have an inherent overhead due to the need for a cyclic prefix. Assuming QPSK modulation, in terms of N. Le. T. and the number of null tones K (all as previously defined), what is the effective data rate of the OFDM system in lab?

Ans: Effective data rate of OFDM system can be given by $\frac{1}{T(N+L_C)} * (N - K) * 2$

Where $\frac{1}{T}$ = Sample Rate = 4M Samples/Sec, N = 64, L_C = 8, k = 4 samples/sec

$$\text{Effective data rate} = \frac{4M}{(64+8)} * (64 - 4) * 2 = 6.67 \text{ M/sec}$$

If N= 1024, L_C = 32

$$\text{Effective data rate} = \frac{4M}{(1024+32)} * (1024 - 4) * 2 = 38.63 \text{ M/sec}$$

(b) Based on your experience in lab, discuss why or why not you might use an OFDM system in a wideband and narrowband system respectively. Include in your discussion some of the benefits/costs of each scenario.

Ans: In narrowband channels, OFDM systems are beneficial because they make efficient use of the available bandwidth. By employing subcarriers, null tones, and a cyclic prefix, OFDM can effectively mitigate the effects of multipath fading, reduce inter-symbol interference (ISI), and enhance overall system performance. However, one drawback of using OFDM in narrowband systems is that it requires a larger Finite Fourier Transform (FFT) size, which increases the complexity and computational demands of the system.

On the other hand, in wideband channels, OFDM is less effective due to its larger bandwidth requirements. This increased bandwidth leads to longer processing times, making the system less efficient. To address this issue, single-carrier modulation can be a more suitable alternative, helping to simplify the system and improve performance.

(c) OFDM systems in general have a large degree of flexibility. Name at least three parameters of OFDM systems which contribute to this flexibility and comment on how they do so.

Ans: Subcarrier Spacing: In OFDM systems, the spacing between subcarriers can be adjusted to adapt to different channel characteristics, such as frequency selectivity. This flexibility allows the system to optimize performance based on the specific conditions of the channel.

Modulation Scheme: OFDM systems are versatile and can support various modulation schemes, including Quadrature Amplitude Modulation (QAM) and Phase Shift Keying (PSK). This capability showcases the adaptability of OFDM, enabling it to meet diverse communication needs.

Guard Interval: The guard interval in OFDM systems plays a crucial role in protecting against inter-symbol interference (ISI) caused by multipath propagation. By adjusting the length of the guard interval, the system can effectively minimize the impact of overlapping signals, ensuring clearer communication.